

When extended operator-norm completeness detects completeness of the range

Autonomous proof-discovery packet for arXiv:1006.1175

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Status. Candidate full negative answer in the source's stated generality, with an exact positive repair; likely valid, human review requested. Model: GPT5.6.

1 Source question

For asymmetric normed spaces (X, p) and (Y, q) , Cobzaş writes $L(X, Y)$ for the continuous linear maps between the symmetrized normed spaces (X, p^s) and (Y, q^s) . The source equips this space with the extended symmetric operator norm

$$\|A\|_{p,q}^* := \sup_{p(x) \leq 1} q^s(Ax) \in [0, \infty].$$

Proposition 3.7 proves that biBanachness of (Y, q) implies completeness of this extended-metric operator space. Remark 3.8(2), on source PDF page 19, asks whether the converse holds, as it does for ordinary operator norms [1]. The exact excerpt is reproduced below.

WE CAN SHOW THAT $L_{p,q}(X, Y)$ IS COMPLETE IN $(L(X, Y), \|\cdot\|)$ IFF (X, p) IS A BANACH SPACE AND (Y, q) IS $\|\cdot\|^*$ -CONVERGENT TO SOME $A \in L(X, Y)$. For $\epsilon = 1$ let $n_0 \in \mathbb{N}$ be such that $\|A_n - A\|^* \leq 1$ for all $n \geq n_0$. Then $\|A\|^* \leq \|A - A_{n_0}\|^* + \|A_{n_0}\|^* \leq 1 + \|A_{n_0}\|^*$, which implies that both A and $-A$ belong to $L_{p,q}(X, Y)$. $\square$

Remarks 3.8. 1. If a sequence (A_n) in $L_{p,q}(X, Y)$ converges to $A \in L(X, Y)$ with respect to the conjugate norm $(\|\cdot\|_{p,q}^*)^{-1}$ of $\|\cdot\|_{p,q}^*$, then $A \in L_{p,q}(X, Y)$.

2. As it is known, in the classical case, the completeness of $(L(X, Y), \|\cdot\|)$ implies the completeness of the normed space Y . We do not know if a similar result holds for the extended norm $\|\cdot\|^*$.

Let n_0 be such that $\|A - A_n\|^* \leq 1$ for all $n \geq n_0$. Then $\|A\|^* \leq \|A - A_{n_0}\|^* + \|A_{n_0}\|^* \leq 1 + \|A_{n_0}\|^* < \infty$, shows that $A \in L_{p,q}(X, Y)$, proving the validity of the assertion from 1.

The answer is no. More precisely, the only obstruction is whether the domain admits a nonzero functional controlled in both p -directions.

2 The domain dichotomy

Define the *bilaterally p -bounded dual*

$$\mathcal{D}_p := \left\{ \varphi \in (X, p^s)^* : b_p(\varphi) := \sup_{p(x) \leq 1} |\varphi(x)| < \infty \right\}.$$

Also let

$$\mathcal{F}_{p,q}(X, Y) := \{A \in L(X, Y) : \|A\|_{p,q}^* < \infty\},$$

the finite-distance component of zero.

Proof intuition

In the classical converse, one fixes a nonzero bounded functional φ on X and embeds Y into $L(X, Y)$ through the rank-one maps $x \mapsto \varphi(x)y$. The same proof works here exactly when φ is bounded on the one-sided unit ball B_p in absolute value. If no such functional exists, Hahn–Banach shows something stronger: no nonzero operator can have finite extended norm. The extended metric then separates every two distinct operators by infinite distance, so all Cauchy sequences are eventually constant.

Theorem 1 (Exact repaired converse). *Let (X, p) and (Y, q) be real asymmetric normed spaces.*

1. *If $\mathcal{D}_p \neq \{0\}$ and $(L(X, Y), \|\cdot\|_{p,q}^*)$ is complete, then (Y, q) is biBanach.*
2. *If $\mathcal{D}_p = \{0\}$, then $\mathcal{F}_{p,q}(X, Y) = \{0\}$ for every (Y, q) . Consequently $(L(X, Y), \|\cdot\|_{p,q}^*)$ is complete for every Y , regardless of whether (Y, q) is biBanach.*

Thus, for a fixed domain (X, p) , operator-space completeness forces range completeness for every asymmetric range precisely when $\mathcal{D}_p \neq \{0\}$.

Proof. Suppose first that $0 \neq \varphi \in \mathcal{D}_p$. Scale $x_0 \in X$ so that $\varphi(x_0) = 1$, and put $b = b_p(\varphi) > 0$. If (y_n) is q^s -Cauchy in Y , define

$$A_n x := \varphi(x) y_n.$$

These maps belong to $L(X, Y)$ because φ is p^s -continuous, and absolute homogeneity of the norm q^s gives the exact identity

$$\|A_n - A_m\|_{p,q}^* = b q^s(y_n - y_m).$$

Hence (A_n) is Cauchy. By the assumed completeness it converges in $\|\cdot\|_{p,q}^*$ to some $A \in L(X, Y)$. Since $|\varphi(x)| \leq b p(x)$, the equality $\varphi(x_0) = 1$ implies $p(x_0) > 0$. Therefore

$$q^s(y_n - A x_0) = q^s((A_n - A)x_0) \leq p(x_0) \|A_n - A\|_{p,q}^* \longrightarrow 0.$$

Every q^s -Cauchy sequence converges, so (Y, q) is biBanach.

Now suppose $\mathcal{D}_p = \{0\}$ and let $A \in \mathcal{F}_{p,q}(X, Y)$. If $A \neq 0$, choose x_1 with $A x_1 \neq 0$. Hahn–Banach in the ordinary normed space (Y, q^s) supplies $g \in (Y, q^s)^*$ with $g(A x_1) \neq 0$. Then $\varphi = g \circ A$ is nonzero and

$$\sup_{p(x) \leq 1} |\varphi(x)| \leq \|g\| \sup_{p(x) \leq 1} q^s(Ax) = \|g\| \|A\|_{p,q}^* < \infty,$$

contradicting $\mathcal{D}_p = \{0\}$. Thus $\mathcal{F}_{p,q}(X, Y) = \{0\}$.

Finally, if (A_n) is Cauchy for the extended metric, then for all sufficiently large m, n one has $\|A_m - A_n\|_{p,q}^* < 1$. The difference belongs to $\mathcal{F}_{p,q}(X, Y) = \{0\}$, so the sequence is eventually constant and hence convergent. $\square$

3 Explicit counterexample

Corollary 1 (Negative answer to Remark 3.8(2)). *There are nonzero X and incomplete Y for which $(L(X, Y), \|\cdot\|^*)$ is complete.*

Proof. Take

$$X = \mathbb{R}, \quad p(t) = u(t) := \max\{t, 0\},$$

so $p^s(t) = |t|$. Every linear functional has the form $\varphi(t) = ct$. If $c \neq 0$, then

$$b_p(\varphi) = \sup_{t \leq 1} |ct| = \infty,$$

because the one-sided unit ball contains the entire negative half-line. Hence $\mathcal{D}_p = \{0\}$.

Let $Y = c_{00}$ with its ordinary ℓ^2 norm, viewed as the symmetric asymmetric norm $q = \|\cdot\|_2$. It is incomplete: the sequence

$$y_N = (1, 1/2, \dots, 1/N, 0, 0, \dots)$$

is Cauchy and its unique limit in ℓ^2 is $(1/n)_{n \geq 1} \notin c_{00}$.

Every $A \in L(X, Y)$ is $A_y(t) = ty$ for some $y \in c_{00}$. Directly, for $y \neq 0$,

$$\|A_y\|_{p,q}^* = \sup_{t \leq 1} \|ty\|_2 = \infty.$$

Thus distinct operators are at infinite distance. Every Cauchy sequence in $L(X, Y)$ is eventually constant, so the operator space is complete although Y is not. $\square$

4 Scope, novelty, and review recommendation

The counterexample uses exactly the source’s definition of asymmetric norm: $p(x) = p(-x) = 0$ forces $x = 0$, but $p(x)$ itself need not be positive for every nonzero x . In particular, u is allowed throughout the source and is one of its basic examples. The result does not assert that the converse fails under an additional T_1 hypothesis. Theorem 1 instead identifies a natural hypothesis— $\mathcal{D}_p \neq 0$ —under which the converse is restored verbatim.

A bounded novelty check through 13 August 2026 searched all four run indexes, the exact arXiv id and title, the exact question wording, and combinations of *extended operator norm*, *completeness*, *biBanach*, and *asymmetric norm*. It found the source, its later book version, and neighboring work on compact and Hahn–Banach extension operators, but no later answer to Remark 3.8(2) and no statement of the dichotomy above. Novelty is therefore plausible but provisional.

Human review should focus on the intended scope of “a similar result,” the use of completeness for an extended metric, and whether T_1 or finite-valued operator-norm hypotheses were tacitly intended despite not appearing in the remark.

References

- [1] S. Cobzaş, *Functional analysis in asymmetric normed spaces*, arXiv:1006.1175v1, 2010.